

Waterfall Project Management Diagram

Project: AI-Powered Second-Hand Trading Platform

Group Name: Group G **Team Members:** Fan Fan, Mahima Hoque Mim

1. Requirements Analysis

Week 1

Define user roles, project objectives, marketplace problems, semantic search needs, AI listing assistant requirements, recommendation goals, and success criteria.

2. System Design

Week 2

Design UI wireframes, database schema, backend API structure, vector search architecture, LLM workflow, recommendation pipeline, and cloud deployment plan.

3. Development

Weeks 3 - 4

Build the responsive frontend, backend API, seller listing parser, buyer search flow, personalized feed, admin dashboard, database, and vector embedding engine.

4. Testing

Week 5

Conduct unit testing, integration testing, usability testing, security checks, AI output validation, search

Key Project Deliverables

- ✓ AI-assisted seller listing workflow
- ✓ Semantic search and ranking engine
- ✓ Personalized recommendation feed
- ✓ Buyer, seller, and admin dashboards
- ✓ Cloud-hosted public application URL
- ✓ Final testing report and maintenance plan

relevance testing, and bug fixing.

5. Deployment

Week 6

Deploy frontend to Vercel/Netlify, backend to Render/Railway, database to Supabase or MongoDB Atlas, then verify public URL and live workflows.

6. Maintenance

Week 7

Monitor platform health, fix issues, improve AI prompts, tune semantic ranking, manage flagged listings, update documentation, and plan future enhancements.

Timeline Flow: Week 1 → Week 2 → Week 3-4 → Week 5 → Week 6 → Week 7